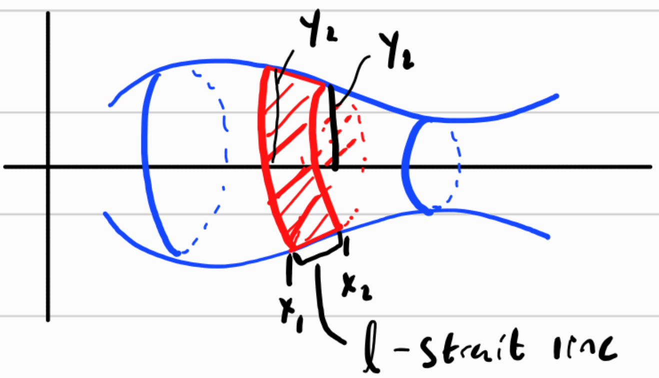
If you recall, a Solid of revolution was some Curve rotated around the x or y axis. The Idea of finding the area was taking the sum of infinitly many discs each of volume $\pi f(x)^2 \Delta x$. This gives the volume as

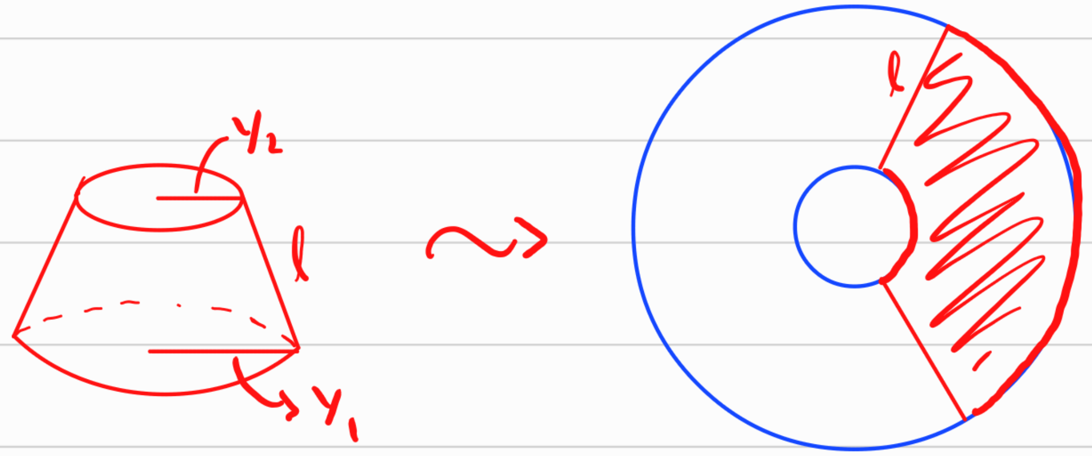
$$V = \pi \int_a^b [f(x)]^2 dx$$

We similary want to find surface area of these objects.

Consider the following picture



Just looking at the red region we have a Cone frustrum,



The surface area of a frustrum is

$$\underline{\underline{A = \pi l (y_1 + y_2)}}$$

and is found from considering a full cone surface area and subtracting the missing smaller cone. The derivation is not that important.

Now, we notice that as we make Δx smaller, $y_1 \to y_2$ and l is just arc length. Taking an infinite sum

Surface area, $S \approx \sum \pi l (y + y) \Delta x = 2\pi \sum s(x)\, y\, \Delta x$

as an integral

$$S = 2\pi \int_a^b \underset{\uparrow\, y}{f(x)} \underset{\uparrow\, s(x)}{\sqrt{1 + \left(\frac{dy}{dx}\right)^2}}\, dx$$

$$\boxed{S = 2\pi \int_a^b y\, ds}$$

* $ds = \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy$ from 8.1

or

$$\boxed{S = 2\pi \int_a^b x\,ds}$$

$$ds = \sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx$$

depending on axis of rotation.

Ex:

Surface area of Sphere

A sphere is found by rotating

$y = \sqrt{r^2-x^2}$ $x \in [-r, r]$ around x-axis.

So,

$$\frac{dy}{dx} = (-2x)\frac{1}{2}(r^2-x^2)^{-1/2} = -\frac{x}{\sqrt{r^2-x^2}}$$

$$\frac{ds}{dx} = \sqrt{1+\left(\frac{-x}{\sqrt{r^2-x^2}}\right)^2} = \sqrt{1+\frac{x^2}{r^2-x^2}} = \sqrt{\frac{r^2-x^2+x^2}{r^2-x^2}}$$

$$ds = \frac{r}{\sqrt{r^2-x^2}}\,dx$$

Then

$$S = 2\pi \int_{-r}^{r} y\,ds = 2\pi \int_{-r}^{r} \sqrt{r^2-x^2}\,\frac{r}{\sqrt{r^2-x^2}}\,dx$$

$$= 2\pi \int_{-r}^{r} r\,dx$$

$$= 2\pi \left[rx \right]_{-r}^{r}$$

$$= 2\pi \left(r^2 - -r^2 \right)$$

$$\boxed{S = 4\pi r^2}$$

Ex: Gaberial's Horn

Rotate $\frac{1}{x}$ on $[1,\infty)$ around x-axis

$$y = \frac{1}{x}$$

$$\frac{dy}{dx} = \frac{-1}{x^2}$$

$$ds = \sqrt{1 + \frac{1}{x^4}}\, dx$$

$$S = \int_1^\infty \frac{1}{x}\sqrt{1+\frac{1}{x^4}}\, dx$$

$$= \int_1^\infty \frac{\sqrt{x^4+1}}{x^3}\, dx$$

$u = x^2 \rightsquigarrow x = \sqrt{u}$

$du = 2x\,dx$

$\downarrow$ $dx = \frac{du}{2\sqrt{u}}$

$$= \frac{1}{2}\int_1^\infty \frac{\sqrt{u^2+1}}{u^2}\, du$$

$u = \tan(\theta)\, d\theta$

$du = \sec^2\theta\, d\theta$

$\sqrt{u^2+1} = \sec\theta$

$$= \frac{1}{2}\int_{\pi/4}^{\pi/2} \frac{\sec^3(\theta)}{\tan^2\theta}\, d\theta$$

$$= \frac{1}{2}\int_{\pi/4}^{\pi/2} \frac{1}{\sin^2\theta\cos\theta}\, d\theta$$

$$= \frac{1}{2}\int_{\pi/4}^{\pi/2} \csc^2\theta \sec\theta \, d\theta$$

$$= \frac{1}{2}\int_{\pi/4}^{\pi/2} (\cot^2\theta + 1)\sec\theta \, d\theta$$

$$= \frac{1}{2}\int_{\pi/4}^{\pi/2} \sec\theta + \cot^2\theta \sec\theta \, d\theta$$

$$= \frac{1}{2}\int_{\pi/4}^{\pi/2} \sec\theta \, d\theta + \int_{\pi/4}^{\pi/2} \frac{\cos^2\theta}{\sin^2\theta}\frac{1}{\cos^2\theta} \, d\theta$$

$$= \frac{1}{2}\ln|\tan\theta + \sec\theta| \Big|_{\pi/4}^{\pi/2} + \frac{1}{2}\int_{\pi/4}^{\pi/2} \cot\theta \csc\theta \, d\theta$$

$$= \frac{1}{2}\ln|\tan\theta + \sec\theta| - \frac{1}{2}\csc\theta \Big|_{\pi/4}^{\pi/2}$$

$$= \frac{1}{2}\ln|\tan\theta + \sec\theta| \Big|_{\pi/4}^{\pi/2} - \frac{1}{2} + \frac{\sqrt{2}}{2}$$

$$= \underbrace{\lim_{t \to \pi/2^-} \frac{1}{2}\ln|\tan t + \sec t|}_{\infty} - \frac{1}{2}\ln|1 + \sqrt{2}| - \frac{1}{2} + \frac{\sqrt{2}}{2}$$

since $\tan t \to \infty$ and $\sec t \to \infty$ as $t \to \pi/2$.

So, $S = \int_1^\infty \frac{1}{x}\sqrt{1 + \frac{1}{x^4}} \, dx$ is divergent so

Gabriel's Horn has infinite surface area, but finite volume (Section 7.8).